

$$1. a) f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

$$\begin{aligned}\mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \cdot e^{-ikx} dx \\ &= \frac{1}{2\pi\sigma} e^{-ik\mu} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2} - ik(x-\mu)} d(x-\mu) \\ &= \frac{1}{2\pi\sigma} \cdot e^{ik\mu} \cdot 2\pi\sigma e^{-\frac{\sigma^2 k^2}{2}} \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{\sigma^2 k^2 + 2ik\mu}{2}}\end{aligned}$$

$$b) f(t) = \sin(\omega_0 t)$$

$$\begin{aligned}\mathcal{F}[f(t)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i} \cdot e^{-i\omega t} dt \\ &= \frac{1}{2i\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i(\omega_0 - \omega)t} dt - \int_{-\infty}^{\infty} e^{-i(\omega_0 + \omega)t} dt \\ &= \frac{1}{2i\sqrt{2\pi}} \cdot 2\pi \cdot (\delta(\omega_0 - \omega) - \delta(\omega_0 + \omega)) \\ &= i\sqrt{\frac{\pi}{2}} (\delta(\omega_0 + \omega) - \delta(\omega_0 - \omega))\end{aligned}$$

$$c) f(x) = e^{-a|x|}$$

$$\begin{aligned}\mathcal{F}[f(x)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a|x|} e^{-ikx} dx \\ &= \frac{1}{\sqrt{2\pi}} \left(\int_{-\infty}^0 e^{(a-ik)x} dx + \int_0^{\infty} e^{-(a+ik)x} dx \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{e^{(a-ik)x}}{a-ik} \Big|_{-\infty}^0 + \frac{e^{-(a+ik)x}}{-(a+ik)} \Big|_0^{\infty} \right) \\ &= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{a-ik} + \frac{1}{a+ik} \right) = \sqrt{\frac{2}{\pi}} \frac{a}{a^2 + k^2}\end{aligned}$$

$$d) f(t) = \delta(t)$$

$$\begin{aligned}\mathcal{F}[f(t)] &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(t) e^{-i\omega t} dt \\ &= \frac{1}{\sqrt{2\pi}} \cdot e^{-i\omega \cdot 0} \\ &= \frac{1}{\sqrt{2\pi}}\end{aligned}$$

$$(2) p \odot q = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p(\tau) q(t+\tau) d\tau$$

$$\text{Since } p(t) = \begin{cases} 1 & -1 \leq t \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$p \odot q = \frac{1}{\sqrt{2\pi}} \int_0^1 q(t+\tau) d\tau$$

$$= \frac{1}{\sqrt{2\pi}} \int_t^{t+1} q(t+\tau) d(t+\tau)$$

if $t+1 < 0$ or $t > 1$ i.e. $t < -1$ or $t > 1$, then $q(t+\tau) = 0$

$$p \odot q = \frac{1}{\sqrt{2\pi}} \int_t^{t+1} 0 d(t+\tau) = 0$$

if $0 \leq t+1 \leq 1$ i.e. $-1 \leq t \leq 0$, then $q(t+\tau) = \begin{cases} 1-t-\tau & \text{if } t+\tau < 1 \\ 0 & \text{otherwise} \end{cases}$

$$p \odot q = \frac{1}{\sqrt{2\pi}} \int_0^{t+1} (1-t-\tau) d(t+\tau)$$

$$= \frac{1}{\sqrt{2\pi}} \left((t+\tau) - \frac{1}{2}(t+\tau)^2 \right) \Big|_0^{t+1}$$

$$= \frac{1}{\sqrt{2\pi}} \left((t+1) - \frac{1}{2}t^2 - t - \frac{1}{2} \right) = \frac{1-t^2}{2\sqrt{2\pi}}$$

if $0 \leq t \leq 1$, then $q(t+\tau) = \begin{cases} 1-t-\tau & \text{if } t+\tau < 1 \\ 0 & \text{otherwise} \end{cases}$

$$p \odot q = \frac{1}{\sqrt{2\pi}} \int_t^1 (1-t-\tau) d(t+\tau)$$

$$= \frac{1}{\sqrt{2\pi}} \left((t+\tau) - \frac{1}{2}(t+\tau)^2 \right) \Big|_t^1$$

$$= \frac{1}{\sqrt{2\pi}} \left(\frac{1}{2} - t + \frac{t^2}{2} \right) = \frac{(t-1)^2}{2\sqrt{2\pi}}$$

To draw a conclusion, $p \odot q = \begin{cases} 0 & t < -1 \\ \frac{1-t^2}{2\sqrt{2\pi}} & -1 \leq t \leq 0 \\ \frac{(1-t)^2}{2\sqrt{2\pi}} & 0 \leq t \leq 1 \\ 0 & t > 1 \end{cases}$

Consider $\langle p \rangle$ and $\langle q \rangle$

Since p and q are both bounded in a limited interval, then their mean on the whole space is 0. i.e. $\langle p \rangle = \langle q \rangle = 0$
 $\langle p \rangle \cdot \langle q \rangle = 0 \neq p \odot q$

Therefore, p and q are correlated.

$$3. \quad p \odot p = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} p(\tau) p(t+\tau) d\tau$$

$$\text{Since } p(t) = \begin{cases} 1 & t \leq 0 \\ 0 & 0 < t < 1 \\ 0 & t > 1 \end{cases}$$

$$\begin{aligned} p \odot p &= \frac{1}{\sqrt{2\pi}} \int_0^1 p(t+\tau) d\tau \\ &= \frac{1}{\sqrt{2\pi}} \int_t^{t+1} p(t+\tau) d(t+\tau) \end{aligned}$$

if $t+1 < 0$ or $t > 1$ i.e. $t < -1$ or $t > 1$, then $p(t+\tau) = 0$

$$p \odot p = \frac{1}{\sqrt{2\pi}} \int_t^{t+1} 0 d(t+\tau) = 0$$

if $0 \leq t+1 \leq 1$ i.e. $-1 \leq t \leq 0$, then $p(t+\tau) = \begin{cases} 1 & \text{if } t+\tau < 1 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} p \odot p &= \frac{1}{\sqrt{2\pi}} \int_0^{t+1} 1 d(t+\tau) \\ &= \frac{1}{\sqrt{2\pi}} (t+\tau) \Big|_0^{t+1} \\ &= \frac{t+1}{\sqrt{2\pi}} \end{aligned}$$

if $0 \leq t \leq 1$, then $p(t+\tau) = \begin{cases} 1 & \text{if } t+\tau < 1 \\ 0 & \text{otherwise} \end{cases}$

$$\begin{aligned} p \odot p &= \frac{1}{\sqrt{2\pi}} \int_t^1 1 d(t+\tau) \\ &= \frac{1}{\sqrt{2\pi}} (t+\tau) \Big|_t^1 \\ &= \frac{1-t}{\sqrt{2\pi}} \end{aligned}$$

$$\text{To draw a conclusion, } p \odot p = \begin{cases} 0 & t < -1 \\ \frac{t+1}{\sqrt{2\pi}} & -1 \leq t \leq 0 \\ \frac{1-t}{\sqrt{2\pi}} & 0 \leq t \leq 1 \\ 0 & t > 1 \end{cases}$$

, and $\langle p \rangle = 0$

Therefore, $p \odot p \neq \langle p \rangle \langle p \rangle$. i.e. p is autocorrelated

$$4. \frac{\partial T}{\partial t} = K \frac{\partial^2 T}{\partial x^2}$$

Using Fourier transforms on x & BHS: $\tau(k, t) = \tilde{F}(T) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T(x, t) e^{-ikx} dx$

$$\frac{\partial \tilde{F}(T)}{\partial t} = K(-ik)^2 \tilde{F}(T) \Rightarrow \frac{\partial \tilde{F}(T)}{\partial t} = -K \cdot k^2 \tilde{F}(T)$$

The general solution for the ODE is:

$$\frac{\partial \tilde{F}(T)}{\tilde{F}(T)} = -K \cdot k^2 dt \Rightarrow \tau(k, t) = C e^{-K \cdot k^2 \cdot t}$$

$$\text{Since } \tilde{F}(T(x, 0)) = \tau(k, 0)$$

$$\Rightarrow \tau(k, t) = \tau(k, 0) \cdot e^{-K \cdot k^2 \cdot t}$$

Using the inverse Fourier transform

$$\begin{aligned} T(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tau(k, t) \cdot e^{ikx} dk \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} T(y, 0) e^{iky} dy \right) \cdot e^{ikx - Kk^2 t} dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} T(y, 0) \left(\int_{-\infty}^{\infty} e^{ik(x-y) - Kk^2 t} dk \right) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} T(y, 0) \sqrt{\frac{\pi}{Kt}} e^{-\frac{(x-y)^2}{4Kt}} dy \\ &= \frac{1}{\sqrt{4\pi Kt}} \int_{-\infty}^{\infty} T(y, 0) e^{-\frac{(x-y)^2}{4Kt}} dy \end{aligned}$$

$$(b) \text{ Since } T(x, 0) = \begin{cases} 0 & |x| > 1 \\ 100 & |x| \leq 1 \end{cases}$$

$$\begin{aligned} T(x, t) &= \frac{1}{\sqrt{4\pi Kt}} \int_{-\infty}^{\infty} T(y, 0) e^{-\frac{(x-y)^2}{4Kt}} dy \\ &= \frac{100}{\sqrt{4\pi Kt}} \int_{-1}^1 e^{-\frac{(x-y)^2}{4Kt}} dy = \frac{100}{\sqrt{4\pi Kt}} \int_{-1-x}^{1-x} e^{-\frac{(x-y)^2}{4Kt}} d(x-y) \\ &= \frac{100}{\sqrt{\pi}} \int_{\frac{-1-x}{\sqrt{4Kt}}}^{\frac{1-x}{\sqrt{4Kt}}} e^{-\left(\frac{x-y}{\sqrt{4Kt}}\right)^2} d\left(\frac{x-y}{\sqrt{4Kt}}\right) \\ &= \frac{100}{\sqrt{\pi}} \cdot (-2\sqrt{e^{-v^2}}) \Big|_{\frac{-1-x}{\sqrt{4Kt}}}^{\frac{1-x}{\sqrt{4Kt}}} = -\frac{200}{\sqrt{\pi}} \cdot \left[\left(\frac{1-x}{\sqrt{4Kt}}\right) \cdot e^{\frac{x^2-2x+1}{4Kt}} - \left(\frac{1+x}{\sqrt{4Kt}}\right) e^{\frac{x^2+2x+1}{4Kt}} \right] \\ &= -\frac{200}{\sqrt{\pi}} \left[\frac{1-x}{2\sqrt{Kt}} \cdot e^{\frac{x^2-2x+1}{4000t}} - \frac{1+x}{2\sqrt{Kt}} e^{\frac{x^2+2x+1}{4000t}} \right] \end{aligned}$$

5.

